

We don't want Layr to be "nice to have."

We want it to be:

If you're in events/exhibitions and you don't use Layr... We're behind.

That means it must solve:

- Creative pain
- Operational pain
- Commercial pain
- Tender pain
- Execution pain

Let's design Layr as a **necessity engine**, not a 3D toy.

- **3D builder**
- **Venue library**
- **Asset warehouse integration**
- **Lighting engine**
- **Cost estimator**
- **AI layout generation**
- **Photo → 3D**
- **Video walkthrough generation**
- **Marketplace**
- **Collaboration layer**

CORE POSITIONING

Layr is not:

"3D mockup software."

Layr is:

The operating system for spatial projects.

If you build it like that, it becomes indispensable.

Core Layer – Spatial Builder

This our your foundation.

Must include:

- **Drag & drop modular assets**

Stage, truss, LED, booths, furniture, lights, flooring, branding panels.

- **Real venue dimensions**
 1. Import hall sizes.
 2. Preset venue templates.
 3. Grid snapping.
 4. Collision detection.

- **Real-world constraints**

1. Height limits.
2. Rigging points.
3. Power access zones.
4. Emergency exits overlay.

If we solve real-world physics, We win.

Venue Intelligence Library

For each venue:

- Accurate 3D model
- Ceiling height
- Pillar positions
- Load restrictions
- Entry doors
- Truck access points

Now agencies don't need site visits for pitching.
That's powerful.

Instant Cost Estimator (Game-Changer)

This makes Layr NECESSARY.

When user builds, It auto-calculates:

1. LED sqm
2. Truss length
3. Carpet area
4. Printing area
5. Structure volume
6. Labor estimation

Optional:

Allow custom cost database per agency.

Now it becomes:

Design + commercial engine.

Huge.

Smart Layout Assistant (AI Layer Later)

Suggest:

1. Optimal booth spacing
2. Best screen placement
3. Sightline improvement
4. Crowd flow suggestions
5. Fire safety compliance warnings
6. Asset generation from prompts and inspirational images

Tender & Presentation Engine

After building, user can:

1. Auto-generate 4K renders
2. Auto-generate top view plan
3. Auto-generate technical drawing PDF
4. Auto-generate bill of quantities summary
5. Auto-generate client presentation deck
6. Now Layr saves 5–10 hours per project.

That's necessity.

Collaboration Layer

Essential.

1. Share view-only link
2. Client comment on 3D scene
3. Version history
4. Compare layout A vs B
5. Internal team notes

Now it replaces WhatsApp chaos.

Global Asset Warehouse Integration

Asset Pipeline System

Layr should support:

- GLB / GLTF
- FBX (converted internally)
- Optimized low-poly web versions
- Auto texture compression

Online Warehouse Integration Options

Integrate APIs or asset ingestion from:

- Sketchfab (with licensing filter)
- TurboSquid (licensed packs)
- CGTrader
- Poly Haven (free PBR assets)

But here's the key:

You do NOT just expose raw warehouses.

You curate + optimize + categorize.

Layr must feel structured, not messy.

Smart Asset Categorization

Assets must be structured like:

- Stage Structures
- LED Systems
- Truss Systems
- Seating Layouts
- Exhibition Booth Modules
- Lighting Fixtures
- Decor
- AV Equipment

Photo → 3D Generation

User snaps photo of:

- Venue
- Stage
- Structure
- Booth

Layr generates:

- 3D spatial approximation
- Depth estimation
- Scene reconstruction

Technically this involves:

- AI depth estimation model
- NeRF-style reconstruction
- Photogrammetry pipeline

Start simple:

Phase 1:

Photo → AI extracts objects → suggests similar 3D models from library.

Phase 2:

Photo → Rough 3D mesh generation.

AI Concept Generator

User types:

“Modern banking summit stage with curved LED wall.”

System generates:

- 3D concept layout
- Suggested assets placed
- Camera angle preview

Technically:

- LLM → interprets prompt
- Layout engine → auto-place modular components
- Asset recommendation engine

This is fundable.

AI spatial auto-layout is strong differentiation.

Video Generation (Walkthrough Engine)

Very important commercially.

Agencies love motion.

Add:

Real-Time Walkthrough Mode

- Smooth camera path generator
- Auto cinematic camera
- Orbit mode
- Fly-through mode
- 4K export

Phase 2:

- AI auto-generates cinematic video
- Smooth easing
- Soft music option + Launch action demo
- Vertical reel export

That alone increases value perception massively.

Lighting Engine

Lighting is what separates amateur from premium.

Layr must have:

Easy Lighting Controls

- Drag & drop spotlight
- Beam angle control
- Intensity slider
- Color picker
- Gobo simulation
- Wash light presets
- LED wall glow control

And critically:

“Auto Light Scene” Button

AI suggests:

- Stage front wash
- Back rim lighting
- Ambient fill
- Highlight panels

Lighting presets:

- Corporate summit
- Gala dinner
- Tech launch
- Concert mode

Lighting makes renders premium.

Real-Time Render vs High-End Render

Two modes:

Fast Mode:

Instant preview (WebGL)

Pro Render:

Cloud-based raytraced high-quality output

Premium feature.

Charge credits for it.

What Makes Layr Truly Dangerous (In a Good Way)

When Layr can:

- Generate concept from text
- Convert photo to spatial suggestion
- Auto place modular booth
- Add lighting automatically
- Generate 4K renders
- Export cinematic video
- Estimate cost
- Generate BOQ
- Estimate Measurements
- Drag and Drop mock up creation

That becomes:

Creative + Operational + Commercial engine.

Asset Marketplace (Future Expansion)

Users can:

1. Buy premium stage templates
2. Buy themed booth designs
3. Buy venue packs
4. Sell their own templates

Now network effects start.

Freelancer Integration (Phase 2/3)

If agency needs help:

They can:

“Hire spatial specialist inside Layr.”

We become ecosystem.

That’s platform thinking.

Analytics & Insights

Show:

1. Most used layouts
2. Most efficient booth sizes
3. Conversion rates per layout
4. Estimated ROI calculator

Now it becomes business intelligence.

Execution-Ready Output

Export:

1. CAD file
2. Fabrication drawings
3. Technical dimensions
4. Material breakdown

Now production teams love it.

White-Label Mode

Large agencies can:

1. Remove Layr branding
2. Present as internal tool
3. Enterprise revenue unlocked.

Multi-Region Expansion

Add:

1. Venue packs per country
2. Regulatory overlays
3. Standard truss systems per market
4. Region-specific materials
5. This supports your Kenya → UAE → India expansion..

The True Killer Feature

If I had to pick ONE feature that makes Layr unavoidable:

Instant cost & material estimator tied to design.
Because agencies suffer there daily.

That converts Layr from:
Creative tool into Revenue engine.
Revenue engines are necessary.

Smart Development Phasing

Phase 1:
Spatial builder + lighting basics + render export.

Phase 2:
Venue library + cost estimation + walkthrough video.

Phase 3:
AI layout suggestions.

Phase 4:
Photo-to-3D + advanced AI concept generation.